TILT SAFETY & ALERT

Curiosity Kickoff

- What happens when the object tilts?
- What happens when it is straight?
- Does it work without touching?

Concept Connection

A Tilt Safety & Alert system detects changes in position or tilt using a tilt sensor and triggers an alert through LEDs or alarms. When the device tilts beyond a set angle, the sensor activates the output, helping in safety applications like accident detection, device protection, and monitoring systems.

1. Tilt Sensor :

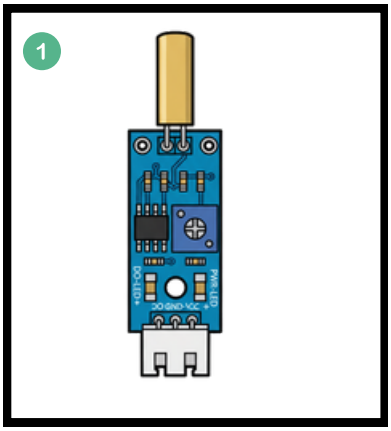
A Tilt Sensor is an electronic device used to detect changes in orientation or angle. When the sensor is tilted, it either opens or closes a circuit, generating a signal. It is commonly used in safety systems, alarms, and motion detection applications.

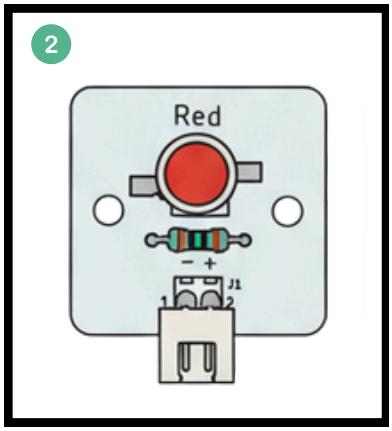
2.Watt SMD LED

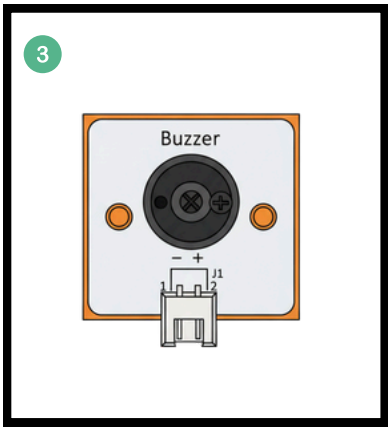
I'm flat and powerful. Just one of me can light up your whole table like a torch.I don't like wasting energy, so I shine super bright without getting too hot,You've already met me — I live in your mobile flashlight!

3.Buzzer

A sound-making device that works when electricity makes a thin metal plate vibrate rapidly inside. The vibration creates buzzing or beeping sounds. Some buzzers can even change frequency to play simple tunes or melodies.







Beginner: Tilt Alert

Class Explanation

A Tilt Alert system uses a tilt sensor to detect changes in orientation or position. When the device tilts beyond a certain angle, it triggers an alert such as an LED or alarm, helping in safety, monitoring, and accident detection applications.

Materials Required

- Arduino with attached shield
- 1 × Tilt sensor
- 1 × Buzzer
- RMC wires
- USB cable

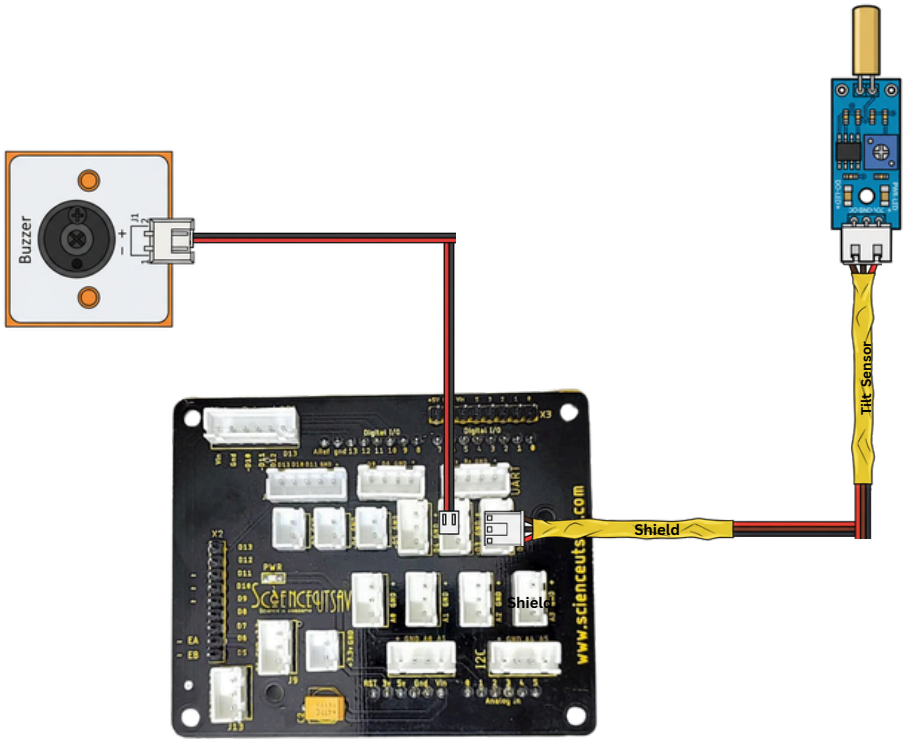
How to Connect

Use RMC wires to connect :

- 1 × Tilt sensor → Arduino pin D2 and GND
- 1 × Buzzer → Arduino pin D3 and GND

Then connect the Arduino to your computer with the USB cable.

“Refer to the link or QR at the back of the book to get the code file for this project.”



Code

```
#define TILT_PIN 2 // Tilt sensor connected to D2
#define BUZZER_PIN 3 // Buzzer connected to D3

void setup() {
 pinMode(TILT_PIN, INPUT);
 pinMode(BUZZER_PIN, OUTPUT);
 digitalWrite(BUZZER_PIN, LOW);
}
void loop() {
 int tiltState = digitalRead(TILT_PIN);

 if (tiltState == HIGH) {
  digitalWrite(BUZZER_PIN, HIGH); // Turn buzzer ON
 } else {
  digitalWrite(BUZZER_PIN, LOW); // Turn buzzer OFF
 }
}
```

Code Explanation

- In this Arduino program, a tilt sensor is connected to pin 2 (TILT_PIN) and a buzzer is connected to pin 3 (BUZZER_PIN).

- The setup() function runs once when the Arduino starts. It sets the tilt sensor pin as INPUT and the buzzer pin as OUTPUT using pinMode(). The buzzer is initially turned OFF using digitalWrite().

- The loop() function runs continuously. It reads the tilt sensor value using digitalRead() and stores it in tiltState.

- If the tilt sensor detects movement (HIGH), the buzzer turns ON. If there is no tilt (LOW), the buzzer turns OFF.

So overall, the buzzer sounds when tilt is detected and stays OFF when there is no tilt.

Try This Yourself Code

This Try This Yourself code keeps the same topic of Tilt Alert Basics but changes the pattern, values, or outputs slightly so students can practice the same idea with more confidence.

Try This Yourself Code

```
#define TILT_PIN 2     // Tilt sensor on D2
#define BUZZER_PIN 3   // Buzzer on D3
#define LED_PIN 5      // LED on D5

void setup() {
  pinMode(TILT_PIN, INPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(LED_PIN, OUTPUT);

  digitalWrite(BUZZER_PIN, LOW);
  digitalWrite(LED_PIN, LOW);
}

void loop() {
  int tiltState = digitalRead(TILT_PIN);

  if (tiltState == HIGH) {
    digitalWrite(BUZZER_PIN, HIGH); // Buzzer ON
    digitalWrite(LED_PIN, HIGH);    // LED ON
  } else {
    digitalWrite(BUZZER_PIN, LOW);  // Buzzer OFF
    digitalWrite(LED_PIN, LOW);     // LED OFF
  }
}
```

Objective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learn

the basic idea of Tilt Alert Basics how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Intermediate: Tilt Safety Alarm

Class Explanation

In this class, students learn A Tilt Safety Alarm system uses a tilt sensor to detect changes in position or angle. When the device tilts beyond a safe limit, it activates an alarm or LED, providing alerts for safety, accident prevention, and equipment protection applications.

Materials Required

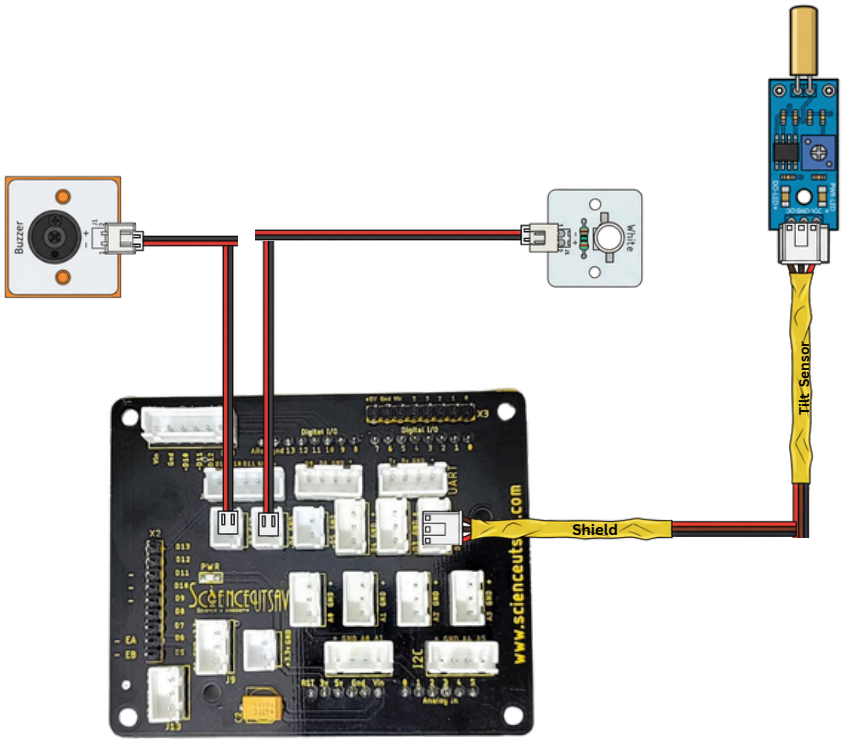
- Arduino with attached shield
- 1 × Tilt sensor
- 1 × white 1 Watt SMD LED on breakout board
- 1 × Buzzer
- RMC wires
- USB cable

How to Connect

- Use RMC wires to connect :
- 1 × Tilt sensor → Arduino pin D2 and GND
- 1 × white 1 Watt SMD LED → Arduino pin D6 and GND
- 1 × Buzzer→ Arduino pin D7 and GND

Then connect the Arduino to your computer with the USB cable.

“Refer to the link or QR at the back of the book to get the code file for this project.”



Code

```
#define TILT_PIN 2  // Tilt sensor connected to D2
#define BUZZER_PIN 7 // Buzzer connected to D7
#define LED_PIN 6   // LED connected to D6

void setup() {
 pinMode(TILT_PIN, INPUT);
 pinMode(BUZZER_PIN, OUTPUT);
 pinMode(LED_PIN, OUTPUT);
 digitalWrite(BUZZER_PIN, LOW);
 digitalWrite(LED_PIN, LOW);
}

void loop() {
 int tiltState = digitalRead(TILT_PIN);

 if (tiltState == HIGH) {
  digitalWrite(BUZZER_PIN, HIGH); // Turn buzzer ON
  digitalWrite(LED_PIN, HIGH);  // Turn LED ON
 } else {
  digitalWrite(BUZZER_PIN, LOW); // Turn buzzer OFF
  digitalWrite(LED_PIN, LOW);  // Turn LED OFF
 }
}
```

Code Explanation

- In this Arduino program, a tilt sensor is connected to pin 2 (TILT_PIN), a buzzer to pin 7 (BUZZER_PIN), and an LED to pin 6 (LED_PIN).

- The setup() function runs once when the Arduino starts. It sets the tilt sensor as INPUT and both the buzzer and LED as OUTPUT using pinMode(). Initially, both the buzzer and LED are turned OFF using digitalWrite().

- The loop() function runs continuously. It reads the tilt sensor value using digitalRead() and stores it in tiltState.

- If tilt is detected (HIGH), both the buzzer and LED turn ON. If there is no tilt (LOW), both turn OFF.

So overall, the Arduino activates the buzzer and LED when tilt is detected and turns them OFF otherwise.

Try This Yourself Code

This Try This Yourself code keeps the same topic of Tilt Safety Alarm System with Arduino but changes the pattern, values, or outputs slightly so students can practice the same idea with more confidence.

Try This Yourself Code

```
#define TILT_PIN 2    // Tilt sensor connected to D2
#define BUZZER_PIN 7  // Buzzer connected to D7
#define LED1_PIN 6    // First LED on D6
#define LED2_PIN 5    // Second LED on D5

void setup() {
  pinMode(TILT_PIN, INPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(LED1_PIN, OUTPUT);
  pinMode(LED2_PIN, OUTPUT);

  digitalWrite(BUZZER_PIN, LOW);
  digitalWrite(LED1_PIN, LOW);
  digitalWrite(LED2_PIN, LOW);
}

void loop() {
  int tiltState = digitalRead(TILT_PIN);

  if (tiltState == HIGH) {
    digitalWrite(BUZZER_PIN, HIGH); // Buzzer ON
    digitalWrite(LED1_PIN, HIGH);   // LED1 ON
    digitalWrite(LED2_PIN, HIGH);   // LED2 ON
  } else {
    digitalWrite(BUZZER_PIN, LOW);  // Buzzer OFF
    digitalWrite(LED1_PIN, LOW);    // LED1 OFF
    digitalWrite(LED2_PIN, LOW);    // LED2 OFF
  }
}
```

Objective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learnctive:

the basic idea of Tilt Safety Alarm System with Arduino how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Advanced: Tilt Safety Alert

Class Explanation

In this class, students learn the A Tilt Safety Alert system uses a tilt sensor to detect changes in position or angle. When the device tilts beyond a safe limit, it activates an alarm or LED, providing alerts for safety, accident prevention, and equipment protection applications.

Materials Required

- Arduino with attached shield
- 1 × Tilt sensor
- 1 × white 1 Watt SMD LED
- 1 × Buzzer
- RMC wires
- USB cable

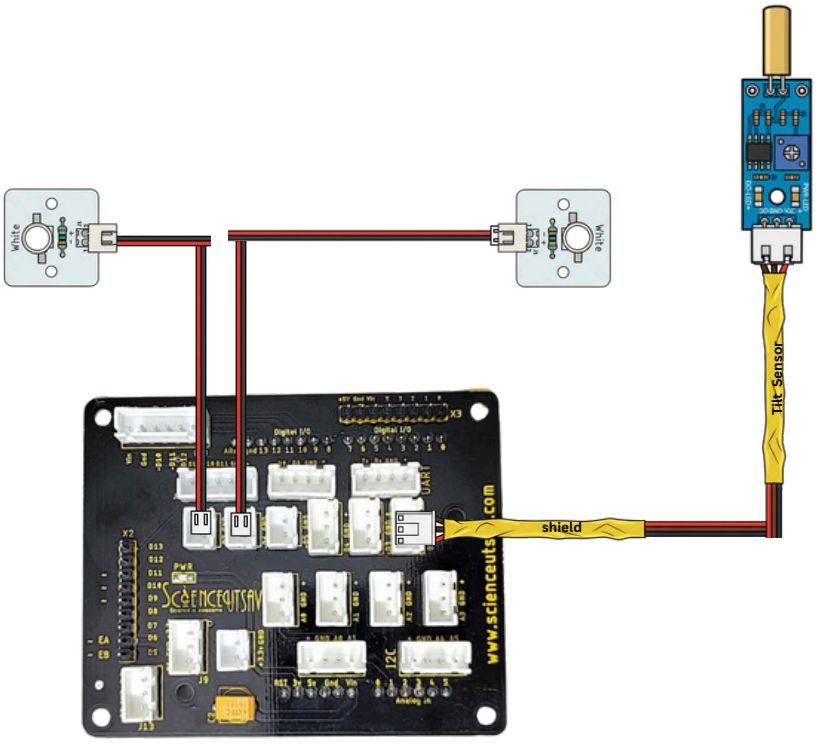
How to Connect

Use RMC wires to connect :

- 1 × Tilt sensor → Arduino pin D2 and GND
- 1 × white 1 Watt SMD LED → Arduino pin D6 and GND
- 1 × Buzzer→ Arduino pin D7 and GND

Then connect the Arduino to your computer with the USB le.

“Refer to the link or QR at the back of the book to get the code file for this project.”



Code

```
#define TILT_PIN 2   // Tilt sensor on D2

#define LED1_PIN 6   // LED1 on D6
#define LED2_PIN 5   // LED2 on D5

void setup() {
  pinMode(TILT_PIN, INPUT);
  pinMode(LED1_PIN, OUTPUT);
  pinMode(LED2_PIN, OUTPUT);

  digitalWrite(LED1_PIN, LOW);
  digitalWrite(LED2_PIN, LOW);
}

void loop() {
  int tiltState = digitalRead(TILT_PIN);

  if (tiltState == HIGH) {
    digitalWrite(LED1_PIN, HIGH); // LED1 ON
    digitalWrite(LED2_PIN, HIGH); // LED2 ON
  } else {
    digitalWrite(LED1_PIN, LOW);  // LED1 OFF
    digitalWrite(LED2_PIN, LOW);  // LED2 OFF
  }
}
```

Code Explanation

- In this Arduino program, a tilt sensor is connected to pin 2 (TILT_PIN), and two LEDs are connected to pins 6 and 5 (LED1_PIN and LED2_PIN).

- The setup() function runs once when the Arduino starts. It sets the tilt sensor as INPUT and both LEDs as OUTPUT using pinMode(). Initially, both LEDs are turned OFF.

- The loop() function runs continuously. It reads the tilt sensor value using digitalRead() and stores it in tiltState.

- If tilt is detected (HIGH), both LEDs turn ON. If there is no tilt (LOW), both LEDs turn OFF.

So overall, both LEDs glow when the tilt sensor is activated and turn OFF when it is not.

Try This Yourself Code

This Try This Yourself code keeps the same topic of Tilt Safety Alarm but changes the pattern, values, or outputs slightly so students can practice the same idea with more confidence.

Try This Yourself Code

```
#define TILT_PIN 2    // Tilt sensor connected to D2
#define BUZZER_PIN 7  // Buzzer connected to D7
#define LED1_PIN 6    // First LED on D6
#define LED2_PIN 5    // Second LED on D5

void setup() {
  pinMode(TILT_PIN, INPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(LED1_PIN, OUTPUT);
  pinMode(LED2_PIN, OUTPUT);

  digitalWrite(BUZZER_PIN, LOW);
  digitalWrite(LED1_PIN, LOW);
  digitalWrite(LED2_PIN, LOW);
}

void loop() {
  int tiltState = digitalRead(TILT_PIN);

  if (tiltState == HIGH) {
    digitalWrite(BUZZER_PIN, HIGH); // Buzzer ON
    digitalWrite(LED1_PIN, HIGH);   // LED1 ON
    digitalWrite(LED2_PIN, HIGH);   // LED2 ON
  } else {
    digitalWrite(BUZZER_PIN, LOW);  // Buzzer OFF
    digitalWrite(LED1_PIN, LOW);    // LED1 OFF
    digitalWrite(LED2_PIN, LOW);    // LED2 OFF
  }
}
```

Obective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learnctive:

the basic idea of Tilt Safety Alarm how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Expert: Smart Tilt Logic

Class Explanation

A Smart Tilt Logic system uses a tilt sensor along with logic circuits to detect position changes and control outputs intelligently. When a tilt is detected, the system processes the signal and activates devices like LEDs or alarms based on predefined logic conditions.

Materials Required

- Arduino with attached shield
- 1 × Tilt sensor
- 1 × Servo motor
- RMC wires
- USB cable

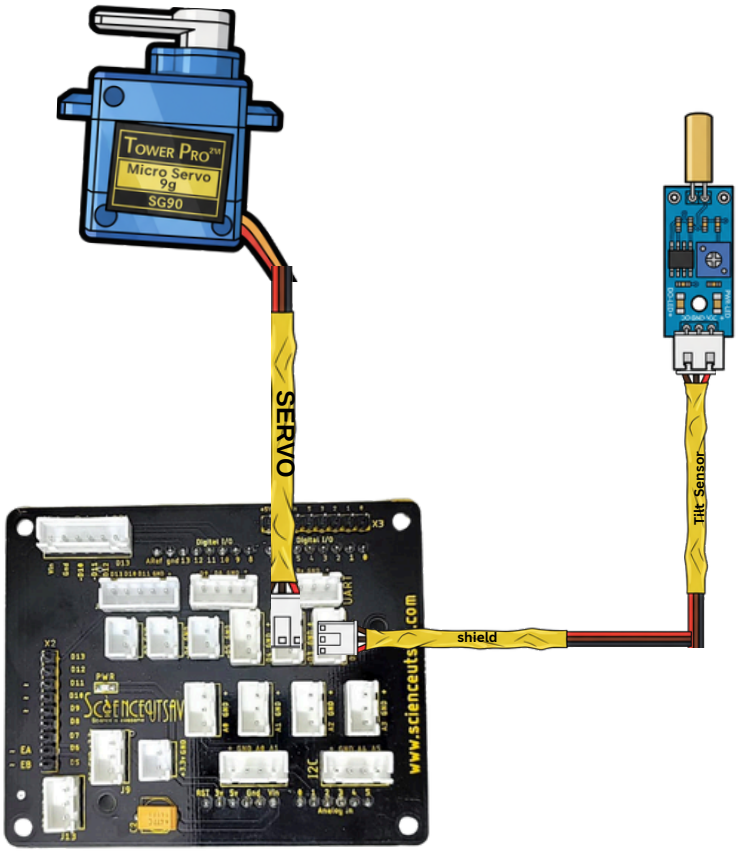
How to Connect

Use RMC wires to connect :

- 1 × Tilt sensor → Arduino pin D2 and GND
- 1 × Servo motor → Arduino pin D3 and GND

Then connect the Arduino to your computer with the USB cable.

“Refer to the link or QR at the back of the book to get the code file for this project.”



Code

```
/// Tilt → Servo demo
// Moves a servo on D3 to 180° when the tilt sensor is HIGH, and back to 0° otherwise.

#include <Servo.h>

#define TILT_PIN 2   // Tilt sensor connected to D2 (digital input)
#define SERVO_PIN 3  // Servo signal connected to D3

Servo tiltServo;

void setup() {
 pinMode(TILT_PIN, INPUT); // Use INPUT_PULLUP if your tilt switch is wired to GND
 tiltServo.attach(SERVO_PIN);
 tiltServo.write(0);    // Start at 0°
}

void loop() {
 int tiltState = digitalRead(TILT_PIN);

 if (tiltState == HIGH) {
  tiltServo.write(180);  // Tilt detected → move to 180°
 } else {
  tiltServo.write(0);   // No tilt → move back to 0°
 }

 delay(20); // small delay to reduce jitter
}
```

Code Explanation

- In this Arduino program, a tilt sensor is connected to pin 2 (TILT_PIN) and a servo motor to pin 3 (SERVO_PIN). The Servo library is used to control the servo.
-
- The setup() function runs once. It sets the tilt sensor as INPUT, attaches the servo to its pin, and sets its initial position to 0°.
-
- The loop() function runs continuously. It reads the tilt sensor using digitalRead().
-
- If tilt is detected (HIGH), the servo moves to 180°. If not (LOW), it returns to 0°. A small delay is added to reduce jitter.
-
- So overall, the servo moves based on the tilt sensor’s position.

Try This Yourself Code

This Try This Yourself code keeps the same topic of Smart Tilt Logic System but changes the pattern, values, or outputs slightly so students can practice the same idea with more confidence.

Try This Yourself Code

```
#include <Servo.h>

#define TILT_PIN 2 // Tilt sensor on D2
#define SERVO_PIN 3 // Servo on D3
#define LED_PIN 6 // LED on D6
#define BUZZER_PIN 5 // Buzzer on D5

Servo tiltServo;

void setup() {
 pinMode(TILT_PIN, INPUT); // Use INPUT_PULLUP if needed
 pinMode(LED_PIN, OUTPUT);
 pinMode(BUZZER_PIN, OUTPUT);

 tiltServo.attach(SERVO_PIN);
 tiltServo.write(0); // Start at 0°

 digitalWrite(LED_PIN, LOW);
 digitalWrite(BUZZER_PIN, LOW);
}

void loop() {
 int tiltState = digitalRead(TILT_PIN);

 if (tiltState == HIGH) {
 tiltServo.write(180); // Move servo
 digitalWrite(LED_PIN, HIGH); // LED ON
 digitalWrite(BUZZER_PIN, HIGH); // Buzzer ON
 } else {
 tiltServo.write(0); // Reset servo
 digitalWrite(LED_PIN, LOW); // LED OFF
 digitalWrite(BUZZER_PIN, LOW); // Buzzer OFF
 }

 delay(20);
}
```

Objective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learn:
the basic idea of Smart Tilt Logic how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Learning Outcomes
After completing this session, students should be able to:

- Understand how tilt sensors detect position changes and trigger alerts to ensure safety and prevent accidents in electronic systems.
- Learn how a tilt sensor works to detect angle changes and activate indicators like LEDs or alarms when tilted.
- Understand how tilt-based systems activate alarms when a device exceeds a safe angle, helping in protection and monitoring applications.
- Learn how tilt input signals are processed using logic circuits to control outputs intelligently based on different tilt conditions.

Try Out Other Things

- use the Try This Yourself code after each main class code before asking students to build a completely new variation
- ask students to point out the exact lines that changed between the first code and the Try This Yourself code
- let students first run the Try This Yourself code code as it is and then change one value, one condition, or one output